Powered by Filecoin Calibration

SafeDrop

The Trustless Dead Man's Switch
Secure your digital legacy on **Web3**

AES-256 Encrypted

XOR Key Splitting

Autonomous Agent

PL_Genesis: Frontiers of Collaboration

The Concept

A trustless system that ensures your digital legacy reaches the right hands, without relying on any single entity.

Client-Side Encryption

Your secret is encrypted **before it leaves your browser** using AES-256-GCM

XOR Key Splitting

One key becomes two shares. **Neither alone reveals anything** — information-theoretically secure

Autonomous Agent

A trustless guardian that **monitors and reveals** your secret when the time comes

1 Encrypt & Split → 2 Distribute Shares → 3 Monitor & Reveal → 4 Beneficiary Decrypts

How It Works

Three-phase trustless workflow

1

Create

- Encrypt with **AES-256-GCM**
- Split key via **XOR**
- Upload to **IPFS/Storacha**
- Register on **Smart Contract**
- Share2 to **Agent**

`Web Crypto API` `Next.js`

2

Monitor

- Timer countdown
- Agent polls **contract** every 30s
- Check in to reset timer
- Early release option available
- Real-time WebSocket updates

`Socket.IO` `Node.js`

3

Recover

- Timer expires → Agent detects
- Agent reveals **Share2** on-chain
- Beneficiary combines both shares
- Fetch ciphertext from **IPFS**
- Decrypt and access secret

`Filecoin FEVM` `React`

The Custodian Agent

An autonomous guardian that never sleeps. No human intervention required — pure cryptographic trust.

Trustless & Autonomous

Encrypted at Rest

AES-256-GCM encryption with HKDF key derivation. Share2 stored securely in SQLite.

Continuous Monitoring

Polls smart contract every 30 seconds for released-but-unrevealed wills.

Auto-Reveal Logic

Automatically calls revealShare() on contract when timer expires without manual intervention.

Real-Time Events

WebSocket broadcasts to connected frontends for instant status updates.

Node.js | Socket.IO | Wagmi/Viem

Tech Stack & Deployment

Production-ready infrastructure on Web3

Blockchain Layer

Filecoin FEVM | Solidity | wagmi v3 | viem v2

Storage Layer

IPFS | Storacha | AES-256-GCM

Application Layer

Next.js 16 | React 19 | Node.js Agent | Socket.IO | Prisma | SQLite

• LIVE Frontend UI

safe-drop-testnet.vercel.app

Vercel

• LIVE Custodian Agent

safedrop-zhu9.onrender.com

Render

Open Source Repository

<> github.com/proskairos/SafeDrop